



**Term Evaluation Theory (Even) Semester Regular Examination February 2026**

Roll no.....

Name of the Course: B.Tech.

Semester: IV

Name of the Paper: *Virtualization and Cloud Computing*

Paper Code: TCS451

Time: 1.5 hour

**Maximum Marks: 50**

**Note:**

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

**Q1. (10 Marks)**

a. Define Cloud Computing and explain its key characteristics. Discuss the characteristics of Cloud Computing as defined by NIST and compare them with general cloud characteristics. CO1

OR

b. Compare full virtualization, partial virtualization and para-virtualization. Discuss their architecture, advantages and limitations. CO2

**Q2. (10 Marks)**

a. Describe the historical developments that led to the emergence of Cloud Computing, including Distributed Systems, Virtualization, Web 2.0, Service-Oriented Computing, and Utility-Oriented Computing. CO1

OR

b. Explain desktop virtualization. Discuss the role of software virtualization and memory virtualization in desktop environments. CO2

**Q3. (10 Marks)**

a. Explain the Cloud Computing Reference Model in detail. Discuss the major components involved in building cloud computing environments. CO1

OR

b. Explain storage virtualization, data virtualization, and network virtualization. Compare their functionalities and real-world use cases. CO2

**Q4. (10 Marks)**

a. Discuss the roles and expectations of different stakeholders in Cloud Computing and analyze how Cloud Computing helps in reducing the Total Cost of Ownership (TCO) compared to traditional on-premises IT infrastructure. CO1

OR

b. Explain Xen para-virtualization, VMware full virtualization and Microsoft Hyper-V. Compare their architectures and use cases in enterprise environments. CO2

**Q5. (10 Marks)**

a. Explain the taxonomy of virtualization techniques. Classify and describe execution virtualization, hardware virtualization, and desktop virtualization. CO1

OR

b. Discuss the advantages and disadvantages of virtualization from the perspectives of performance, cost, security, and management. CO1, CO2